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REFRIGERATOR APPLIANCE ANTI-RACKING SLIDING DRAWER

Abstract

A slidably mountable drawer assembly for a refrigerator appliance is provided. The drawer assembly includes a drawer defining a chamber between a first side and a second side. The drawer assembly also includes a first rail, and a second rail. A drawer flange extends laterally from both the first side and second side of the drawer. The drawer flange is configured to engage with both the first rail and the second rail. The drawer comprises one or more tabs extending from the drawer flange on at least one of the first side and second side of the drawer. The one or more tabs are each configured to engage a respective one of a first slot in the first rail or a second slot in the second rail.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present disclosure is related generally to refrigerator appliances, and more particularly to systems for anti-racking sliding drawers in a refrigerator appliance.

BACKGROUND OF THE INVENTION

[0002] Refrigerator appliances generally include a cabinet that defines a chilled chamber. A wide variety of food items may be stored within the chilled chamber. The low temperature of the chilled chamber relative to ambient atmosphere assists with increasing a shelf life of the food items stored within the chilled chamber.

[0003] However, refrigerator sliding drawers, such as produce drawers, vegetable drawers and freezer drawers, are typically mounted to the refrigerators case through a pair of independent drawer slides. These slides are generally uncontrolled with respect to each other, such that the front of the drawer may deviate out of parallel with respect to the case during drawer actuation. This deviation is undesirable and may lead to user belief of a low degree of craftsmanship, but further may negatively affect the mounting and functionality of the drawers.

[0004] Accordingly, a refrigerator with anti-racking sliding drawers therein would be advantageous.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one example embodiment, a refrigerator appliance is provided. The refrigerator appliance defines a vertical direction, a lateral direction, and a transverse direction. The vertical, lateral, and transverse directions are mutually perpendicular. The refrigerator appliance includes a cabinet defining a food storage chamber. The food storage chamber extends between a top portion and a bottom portion along the vertical direction, between a first side portion and a second side portion along the lateral direction, and between a front portion and a back portion along the transverse direction. A first support member and a second support member are disposed proximate the back portion of the food storage chamber for providing a plurality of shelf mounting positions. The refrigerator appliance also includes a slidably mountable drawer assembly. The slidably mountable drawer assembly includes a drawer. The drawer includes an exterior, an interior, and a chamber defined in the interior of the drawer. The chamber of the drawer defines a food storage compartment. A drawer flange extends laterally from both a first side of the drawer and a second side of the drawer. A first tab extends from the drawer flange on the first side of the drawer, and a second tab extends from the drawer flange on the second side of the drawer. The slidably mountable drawer assembly also includes a first rail extending along the transverse direction on the first side of the exterior of the drawer. The first rail comprising a first rail flange defining a first slot, the first slot configured to receive the first tab. A second rail extends along the transverse direction on the second side of the exterior of the drawer. The second side is opposite the first side. The second rail includes a second rail flange defining a second slot. The second slot is configured to receive the second tab. The drawer flange is configured to engage with both the first rail and the second rail. The first tab is configured to engage the first rail and the second tab is configured to engage the second rail.

[0007] In a second example embodiment, a refrigerator appliance is provided. The refrigerator appliance defines a vertical direction, a lateral direction, and a transverse direction. The vertical, lateral, and transverse directions are mutually perpendicular. The refrigerator appliance includes a cabinet defining a food storage chamber. The food storage chamber extends between a top portion and a bottom portion along the vertical direction, between a first side portion and a second side portion along the lateral direction, and between a front portion and a back portion along the transverse direction. The refrigerator appliance also includes a slidably mountable drawer

assembly. The slidably mountable drawer assembly includes a drawer that includes an exterior, an interior, and a chamber defined in the interior of the drawer. The chamber of the drawer defines a food storage compartment. A first rail extends along the transverse direction on a first side of the exterior of the drawer. The first rail includes a first rail flange with a first slot defined in the first rail flange. A second rail extends along the transverse direction on a second side of the exterior of the drawer. The second side is opposite the first side. The second rail includes a second rail flange with a second slot defined in the second rail flange. A drawer flange extends laterally from both the first side and second side of the drawer. The drawer flange is configured to engage with both the first rail and the second rail. The drawer comprises one or more tabs extending from the drawer flange on at least one of the first side and second side of the drawer. The one or more tabs are each configured to engage a respective one of the first slot in the first rail or the second slot in the second rail.

[0008] In another example embodiment, a slidably mountable drawer assembly for a refrigerator appliance is provided. The drawer assembly includes a drawer defining a vertical direction, a lateral direction, and a transverse direction. The vertical, lateral, and transverse directions are mutually perpendicular. The drawer includes an exterior, an interior, and a chamber defined in the interior of the drawer. The chamber of the drawer defines a food storage compartment. A first rail extends along the transverse direction on a first side of the exterior of the drawer. The first rail includes a first rail flange with a first slot defined in the first rail flange. A second rail extends along the transverse direction on a second side of the exterior of the drawer. The second side is opposite the first side. The second rail includes a second rail flange with a second slot defined in the second rail flange. A drawer flange extends laterally from both the first side and second side of the drawer. The drawer flange is configured to engage with both the first rail and the second rail. The drawer comprises one or more tabs extending from the drawer flange on at least one of the first side and second side of the drawer. The one or more tabs are each configured to engage a respective one of the first slot in the first rail or the second slot in the second rail.

[0009] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0011] FIG. 1 provides a front view of a refrigerator appliance according to aspects of the present disclosure.

[0012] FIG. 2 provides a front perspective view of the refrigerator appliance of FIG. 1 with refrigerator doors and a freezer door shown in an open configuration according to aspects of the present disclosure.

[0013] FIG. 3 provides a perspective view of a first position of a drawer assembly slidably mounted to a shelf, such as a shelf of the refrigerator appliance of FIG. 1 according to aspects of the present disclosure.

[0014] FIG. 4 provides a perspective view of the drawer assembly of FIG. 3 in a second position.

[0015] FIG. 5 provides a perspective view of an example embodiment of a drawer assembly according to aspects of the present disclosure.

[0016] FIG. 6 provides a perspective section view of a portion of the drawer assembly of FIG. 5.

[0017] FIG. 7 provides a top, schematic view of the drawer assembly of FIG. 5 according to aspects of the present disclosure.

[0018] FIG. 8 provides a front, section view of the drawer assembly of FIG. 5 according to aspects of the present disclosure.

[0019] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0020] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0021] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0022] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0023] References to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0024] FIG. 1 provides a front view of a refrigerator appliance **100** according to an example embodiment of the present subject matter. Refrigerator appliance **100** extends between a top

portion **101** and a bottom portion **102** along a vertical direction V. Refrigerator appliance **100** also extends between a first side portion **105** and a second side portion **106** along a lateral direction L. A transverse direction T (FIG. 2) is defined perpendicular to the vertical and lateral directions V, L. Accordingly, vertical direction V, lateral direction L, and transverse direction T are mutually perpendicular.

[0025] Refrigerator appliance **100** includes a housing or cabinet **120** defining a volume **121** including at least one food storage chamber defined therein. In the illustrated example embodiment, cabinet **120** defines multiple food storage chambers, e.g., an upper fresh food chamber **122** and a lower freezer chamber **124** arranged below the fresh food chamber **122** on the vertical direction V. As such, refrigerator appliance **100** is generally referred to as a bottom mount refrigerator. In this example embodiment, cabinet **120** also defines a mechanical compartment (not shown) for receipt of a sealed cooling system (not shown). It will be appreciated that the present subject matter can be used with other types of refrigerators (e.g., side-by-sides), freezer appliances, and/or other types of appliances more generally. Consequently, the description set forth herein is for example purposes only and is not intended to limit the scope of the present subject matter in any aspect.

[0026] Refrigerator appliance **100** includes refrigerator doors **126**, **128** that are rotatably hinged to an edge of cabinet **120** for accessing fresh food chamber **122**. It should be noted that while doors **126**, **128** are depicted in a “French door” configuration, any suitable arrangement or number of doors is within the scope and spirit of the present subject matter. A freezer door **130** is arranged below refrigerator doors **126**, **128** for accessing freezer chamber **124**.

[0027] Operation of refrigerator appliance **100** can be regulated by a controller **134** that is operatively coupled to a user interface panel **136**. Panel **136** provides selections for user manipulation of the operation of refrigerator appliance **100**. In response to user manipulation of user interface panel **136**, controller **134** operates various components of refrigerator appliance **100**. Controller **134** may include a memory and one or more processors, microprocessors, CPUs, or the like, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with operation of refrigerator appliance **100**. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor.

[0028] Controller **134** may be positioned in a variety of locations throughout refrigerator appliance **100**. In the illustrated embodiment, controller **134** is located within door **126**. In such an embodiment, input/output (“I/O”) signals may be routed between the controller and various operational components of refrigerator appliance **100**. In one embodiment, user interface panel **136** may represent a general purpose I/O (“GPIO”) device or functional block. The user interface **136** may include input components, such as one or more of a variety of electrical, mechanical, or electro-mechanical input devices including rotary dials, push buttons, and touch pads. User interface **136** may include a display component, such as a digital or analog display device designed to provide operational feedback to a user. The user interface **136** may be in communication with controller **134** via one or more signal lines or shared communication busses.

[0029] FIG. 2 provides a front, perspective view of refrigerator appliance **100** having refrigerator doors **126**, **128** in an open position to reveal the interior of fresh food chamber **122**. Additionally, freezer door **130** is shown in an open position to reveal the interior of freezer chamber **124**. As shown more clearly in FIG. 2, refrigerator appliance **100** extends in the transverse direction T between a front portion **108** and a back portion **110**.

[0030] As shown in FIG. 2, for this example embodiment, fresh food chamber **122** of refrigerator appliance **100** includes a first support member **150** and a second support member **152** disposed proximate the back portion **110**, e.g., mounted to a rear wall **138** of cabinet **120**. The first and second support members **150** and **152** are oriented generally along the vertical direction V. One or

more adjustable shelves **156** may be mounted to the first and second support members **150** and **152**. First and second support members **150** and **152** structurally support one or more adjustable shelves **156**. Moreover, first and second support members **150** and **152** structurally support the weight of other components, e.g., the additional weight from a slidably mountable drawer assembly **200** (FIG. 3). First and second support members **150** and **152** can be made of any suitable structural material. For example, in some embodiments, first and second support members **150** and **152** may be made of steel.

[0031] In the embodiment illustrated in FIG. 2, four (4) adjustable shelves **156** are mounted within fresh food chamber **122** and are arranged in two columns and two rows as shown. Adjustable shelves **156** may be selectively positioned by a user in different shelf mounting positions within fresh food chamber **122**. For instance, one adjustable shelf **156** could be removed from its position and moved upward or downward along the vertical direction V or moved from a position proximate first side **105** to a position proximate second side **106** of refrigerator appliance **100** along the lateral direction L. Adjustable shelves **156** can also be removed from refrigerator appliance **100**. For example, if storage room is needed for a particularly tall pot, adjustable shelves **156** can be removed from refrigerator appliance **100** and stowed elsewhere. Although four (4) adjustable shelves **156** are depicted in FIG. 2, more or less than four (4) adjustable shelves **156** can be provided in refrigerator appliance **100**.

[0032] As shown in FIG. 2, each adjustable shelf **156** may include a first bracket **158** and a second bracket **160** for mounting the shelf **156** to the first and second support members **150** and **152** in one of the shelf mounting positions. For example, the shelf **156** may extend between the first bracket **158** and the second bracket **160** along the lateral direction L. As is generally understood in the art, the first and second support members **150** and **152** may include a plurality of openings and the first and second brackets **158** and **160** may include hooks configured to selectively engage and disengage the openings such that the shelves **156** can be positioned as desired.

[0033] FIGS. 3 and 4 provide perspective views of a drawer assembly **200** slidably mounted to shelf **156**. As illustrated, the slidably mountable drawer assembly **200** includes a first rail **202** and a second rail **204** with a drawer **206** therebetween. As shown in FIGS. 3 and 4, only first rail **202** is visible; however, a person having ordinary skill in the art will understand that a second mirrored rail **204** is also provided. In general, drawer **206** may include an exterior **212**, such as an exterior surface, an interior **214**, such as an interior surface, and a chamber **216** defined by interior **214** of drawer **206**. For example, chamber **216** of drawer **206** may define a food storage compartment generally configured to receive food items for storage within refrigerator appliance **100**. In general, drawer **206** may be mounted to rails **202**, **204**, as will be described in further detail hereinbelow. More specifically, drawer **206** may extend between a first side **218** and a second side **220** along the lateral direction L, where the second side **220** is opposite the first side **218**. The first rail **202** may extend along the transverse direction T at the first side **218** of the drawer **206** and the second rail **204** may extend along the transverse direction T at the second side **220** of the drawer **206**. In the present example embodiment, the first rail **202** and the second rail **204** of the drawer assembly **200** may be configured for slidably mounting to the first bracket **158** and the second bracket **160**, respectively, proximate a bottom surface **154** of the shelf **156**. Drawer **206** of the slidably mountable drawer assembly **200** may be slidable relative to the shelf **156**, e.g., along the transverse direction T. For example, FIG. 3 illustrates drawer **206** in a first (closed) position and FIG. 4 illustrates drawer **206** in a second (open) position, whereby in the open position chamber **216** of drawer **206** may be exposed, such as for a user to retrieve or store food items.

[0034] Turning now to FIG. 5, illustrated is another example embodiment of drawer assembly **200**. The present example embodiment may be a freezer drawer configured to translate generally along the transverse direction T within freezer chamber **124** and at least partially outside of the freezer chamber **124**, e.g., when in an open position. More specifically, first and second rails **202**, **204** may generally be configured to couple to the interior of freezer chamber **124**, or couple to brackets (not

shown) coupled to the interior of freezer chamber **124**. Drawer assembly **200** may extend between a front side **210** (FIG. 7) and a back side **211** (FIG. 7) along the transverse direction T and include a drawer flange **208** extending laterally from both the first side **218** and second side **220** of drawer **206**. Drawer flange **208** may generally be configured to engage with both the first rail **202** and the second rail **204**, respectively, as will be described in further detail hereinbelow.

[0035] Referring now to FIG. 6, illustrated is an upward perspective view of a corner of drawer assembly **200**, specifically illustrating drawer flange **208** engaging first rail **202**. In general, drawer **206** may include a tab **230** extending, e.g., downward generally along the vertical direction V, from drawer flange **208** on each of the first side **218** and second side **220** of drawer **206**. For example, each tab extending from the drawer flange **208** may engage either the first rail **202** or the second rail **204**, respectively. In particular, each of first rail **202** and second rail **204** may include a rail flange extending therefrom, such as a first rail flange **203** of first rail **202** and a second rail flange **205** (FIG. 7) of second rail **204**. Rail flange **203** may generally define a rail slot **240** configured to receive the tab **230** from drawer **206**. In general, tab **230** may be formed with drawer **206**, such as through injection molding or additive manufacturing, such that the drawer **206** and tab **230** are a single piece construction, thus advantageously reducing parts, such as fasteners, as used in traditional assemblies.

[0036] Turning now to FIG. 7, illustrated is a schematic view of an example embodiment of drawer assembly **200**, and more specifically of example tabs of drawer **206** engaging respective slots of first rail **202** and second rail **204**. In particular, in some example embodiments, drawer **206** may include at least three tabs extending from drawer flange **208** of drawer **206**. For example, the first side **218** may include a first tab **230** and a third tab **232**, and the second side **220** may include a second tab **234**. The present illustrated embodiment also includes a fourth tab **236** extending from drawer flange **208** at the second side **218** of the drawer **206**.

[0037] In general, each tab **230**, **232**, **234**, **236** may include a respective protrusion within each respective slot **240**, **242**, **244**, **246**. In particular, the first tab **230** may include a first protrusion **231** extending in the lateral direction L, such that the first protrusion **231** may be configured to engage respective slot **240** of the first rail **202** along the lateral direction L. For example, first protrusion **231** may engage, e.g., push against or contact, slot **240** along the lateral direction L such as to reduce undesired lateral movement of drawer **206**. The third tab **232** may include a third protrusion **233** generally extending in both of the lateral direction L and the transverse direction T, such that the third protrusion **233** may be configured to engage respective slot **242** of the first rail **202** in both the lateral direction L and the transverse direction T. For example, third protrusion **233** may engage, e.g., push against or contact, slot **242** along the lateral direction L and the transverse direction T such as to reduce undesired lateral and transverse movement of drawer **206**.

Accordingly, third protrusion **233** may be otherwise referred to as a four-way locator.

[0038] Moreover, the second tab **234** may include a second protrusion **235** extending in the transverse direction T, such that the second protrusion **235** may be configured to engage respective slot **244** of the second rail **204** along the transverse direction T. For example, second protrusion **235** may engage, e.g., push against or contact, slot **244** along the transverse direction T such as to reduce undesired transversal movement of drawer **206**. In the present example embodiment, first tab **230** and second tab **234**, more specifically first protrusion **231** and second protrusion **235**, may define gaps, such as first gap **241** and second gap **245** from the respective slots **240**, **244**. For example, first protrusion **231** may be loosely fit, e.g., without contacting/engaging, within respective slot **240** with respect to the transverse direction T, and second protrusion **235** may be loosely fit within respective slot **244** with respect to the lateral direction L. Accordingly, each of first protrusion **231** and second protrusion **235** may be otherwise referred to as a two-way locator.

[0039] Additionally, the fourth tab **236** may include a fourth protrusion **237** generally configured to loosely fit, e.g., without contacting/engaging, respective slot **246** of the second rail **204**. Fourth protrusion **237** may provide positional support for drawer **206**, generally aiding assembly of drawer

206 to first and second rails **202**, **204**. Accordingly, first and second rails **202**, **204** may operate in sync, e.g., to provide anti-racking and/or anti-canting control, along four horizontal degrees of freedom. For example, front-to-back translation along the transverse direction as well as side-to-side translation along the lateral direction of drawer **206** may be controlled via each tab **230**, **232**, **234**, **236** engaging each respective slot **240**, **242**, **244**, **246**.

[0040] Additionally, the respective protrusions **231**, **233**, **235**, **237** of tabs **230**, **232**, **234**, **236** may control undesired horizontal rotation of drawer **206**, while asserting mutual, e.g., simultaneous, front-to-back translation of the first and second rails **202**, **204**. In other words, third tab **232** may be a four-way locator and first tab **230** may be a side-to-side two-way locator. First tab **230** and third tab **232** may be paired on one side, e.g., the first side **218**, of drawer **206** with first rail **202**. In other example embodiments, first tab **230** and third tab **232** may be paired on the second side **220** of drawer **206** with second rail **204**. Any arrangement of first tab **230** and third tab **232** may be suitable such that the pair (both first tab **230** and third tab **232**) are positioned on the same side of drawer **206**. Specifically, the combination of first tab **230** and third tab **232** controls three degrees of freedom of drawer **206**, e.g., translation and rotation with respect to first rail **202**, and the second rail **204** may have a single front-to-back two-way locator, e.g., second tab **234**, to control translation of second rail **204** to first rail **202** (the fourth degree of freedom). As such, in some example embodiment, second tab **234** may be provided at any location along second side **220** of drawer **206**. The fourth tab **236**, as seen in FIG. 7 may generally maintain clearance in both directions (lateral and transverse) and may not further constrain the location of drawer **206** on first and second rails **202**, **204**.

[0041] Turning now to FIG. 8, provided is a front, section view illustrating drawer flange **208** on second side **220** of drawer **206** engaging second rail **204**. More specifically, illustrated is an example embodiment of second tab **234** engaging within respective slot **244** of second rail **204**. As stated above, four horizontal degrees of freedom may be controlled the respective protrusions **231**, **233**, **235**, **237** of tabs **230**, **232**, **234**, **236** engaging the respective slots **240**, **242**, **244**, **246**, however, in some example embodiments, out-of-plane (such as the horizontal plane created by the lateral and transverse direction) torsional forces may cause drawer **206** to twist or plastically deform. For example, torsional forces may occur in scenarios where first rail **202** and second rail **204** apply unequal forces, e.g., such as a user pulling a handle of the drawer **206** off-center or a user pushing or twisting the drawer from side-to-side.

[0042] Accordingly, in order to reduce twisting (and thus increase control of the mutual translation of drawer **206** with respect to first and second rail **202**, **204**), drawer flange **208** may be retained to first and second rails **202**, **204**. In particular, the tabs, such as tabs **230**, **232**, **234**, **236**, extending from the drawer flange **208** may include a snap **250** configured to engage, in the vertical direction V, a bottom side **252** of the rail flange **203**. In general, snap **250** may resist torsion of drawer **206**. For example, snap **250** may be an integrated plastic snap, either integrated with tabs **230**, **232**, **234**, **236**, or in some example embodiments, integrated with drawer flange **208** such that snaps **250** extend from drawer flange **208** independently from tabs **230**, **232**, **234**, **236**. As such, in the present illustrated embodiment, snap **250** may engage, in the vertical direction V, with bottom side **252** of the second rail flange **205** such that torsion out of the horizontal plane created by the lateral and transverse direction may be reduced and/or controlled.

[0043] As may be seen from the above, provided may be locating and retention structures for a sliding drawer assembly in a refrigerator appliance. A drawer of drawer assembly may be slidable front-to-back via a pair of rails which may be otherwise independent of each other. The drawer may include a mounting flange around the top perimeter of the drawer which mounts on top of the rails. Multiple tabs may be provided on the underside of the bin mounting flange that protrude downward into corresponding slots within the rails. The tabs may provide control over four-degrees of motion, such as front-to-back and side-to-side translation of the drawer, horizontal rotation of the drawer, and mutual front-to-back translation of the rails. Accordingly, the provided structures for a sliding

drawer assembly drawer may advantageously reduce parts count, parts cost, assembly complexity, and labor and increase perceived product quality from the user compared to traditional systems for sliding drawer assemblies in refrigerator appliances.

[0044] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the refrigerator appliance comprising: a cabinet defining a food storage chamber, the food storage chamber extending between a top portion and a bottom portion along the vertical direction, a first side portion and a second side portion along the lateral direction, and a front portion and a back portion along the transverse direction; a first support member and a second support member disposed proximate the back portion of the food storage chamber for providing a plurality of shelf mounting positions; and a slidably mountable drawer assembly comprising: a drawer comprising: an exterior, an interior, a chamber defined in the interior of the drawer, the chamber of the drawer defining a food storage compartment, a drawer flange extending laterally from both a first side of the drawer and a second side of the drawer, a first tab extending from the drawer flange on the first side of the drawer, and a second tab extending from the drawer flange on the second side of the drawer; a first rail extending along the transverse direction on the first side of the exterior of the drawer, the first rail comprising a first rail flange defining a first slot, the first slot configured to receive the first tab; and a second rail extending along the transverse direction on the second side of the exterior of the drawer, the second side opposite the first side, the second rail comprising a second rail flange defining a second slot, the second slot configured to receive the second tab; wherein the drawer flange is configured to engage with both the first rail and the second rail, wherein the first tab is configured to engage the first rail and the second tab is configured to engage the second rail.

2. The refrigerator appliance of claim 1, wherein the drawer comprises at least three tabs, wherein the first side comprises the first tab and a third tab, and the second side comprises the second tab.

3. The refrigerator appliance of claim 2, wherein the first tab comprises a first protrusion extending in the lateral direction, the first protrusion configured to engage the first slot of the first rail in the lateral direction.

4. The refrigerator appliance of claim 2, wherein the third tab comprises a second protrusion extending in both the lateral direction and the transverse direction, the second protrusion configured to engage a third slot of the first rail in both the lateral direction and the transverse direction.

5. The refrigerator appliance of claim 2, wherein the second tab comprises a third protrusion extending in the transverse direction, the third protrusion configured to engage the second slot of the second rail in the transverse direction.

6. The refrigerator appliance of claim 2, further comprising a fourth tab extending from the second side of the drawer, the fourth tab comprising a fourth protrusion configured to loosely fit within a fourth slot of the second rail.

7. The refrigerator appliance of claim 1, wherein the first tab extending from the drawer flange

comprises a first snap configured to engage, in the vertical direction, a bottom side of the first rail flange.

8. A refrigerator appliance defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the refrigerator appliance comprising: a cabinet defining a food storage chamber, the food storage chamber extending between a top portion and a bottom portion along the vertical direction, a first side portion and a second side portion along the lateral direction, and a front portion and a back portion along the transverse direction; a slidably mountable drawer assembly comprising: a drawer comprising an exterior, an interior, and a chamber defined in the interior of the drawer, the chamber of the drawer defining a food storage compartment; a first rail extending along the transverse direction on a first side of the exterior of the drawer, the first rail comprising a first rail flange with a first slot defined in the first rail flange; a second rail extending along the transverse direction on a second side of the exterior of the drawer, the second side opposite the first side, the second rail comprising a second rail flange with a second slot defined in the second rail flange; and a drawer flange extending laterally from both the first side and second side of the drawer, the drawer flange configured to engage with both the first rail and the second rail, wherein the drawer comprises one or more tabs extending from the drawer flange on at least one of the first side and second side of the drawer, wherein the one or more tabs are each configured to engage a respective one of the first slot in the first rail or the second slot in the second rail.

9. The refrigerator appliance of claim 8, wherein the one or more tabs of the drawer comprises at least three tabs, the at least three tabs comprising a first tab and a third tab extending from the drawer flange on the first side of the drawer and a second tab extending from the drawer flange on the second side of the drawer.

10. The refrigerator appliance of claim 9, wherein the first tab comprises a first protrusion extending in the lateral direction, the first protrusion configured to engage the first slot of the first rail in the lateral direction.

11. The refrigerator appliance of claim 9, wherein the third tab comprises a second protrusion extending in both the lateral direction and the transverse direction, the second protrusion configured to engage a third slot of the first rail in both the lateral direction and the transverse direction.

12. The refrigerator appliance of claim 9, wherein the second tab comprises a third protrusion extending in the transverse direction, the third protrusion configured to engage the second slot of the second rail in the transverse direction.

13. The refrigerator appliance of claim 9, further comprising a fourth tab extending from the drawer flange on the second side of the drawer, the fourth tab comprising a fourth protrusion configured to loosely fit within a fourth slot defined in the second rail flange of the second rail.

14. The refrigerator appliance of claim 8, wherein each tab of the one or more tabs extending from the drawer flange comprises a snap configured to engage, in the vertical direction, a bottom side of one of the first rail flange or the second rail flange.

15. A slidably mountable drawer assembly for a refrigerator appliance, the drawer assembly comprising: a drawer defining a vertical direction, a lateral direction, and a transverse direction, the vertical, lateral, and transverse directions being mutually perpendicular, the drawer comprising an exterior, an interior, and a chamber defined in the interior of the drawer, the chamber of the drawer defining a food storage compartment; a first rail extending along the transverse direction on a first side of the exterior of the drawer, the first rail configured to couple to the refrigerator appliance, the first rail comprising a first rail flange defining a first slot; a second rail extending along the transverse direction on a second side of the exterior of the drawer, the second side opposite the first side, the second rail configured to couple to the refrigerator appliance, the second rail comprising a second rail flange defining a second slot; and a drawer flange extending laterally from both the first side and second side of the drawer, the drawer flange configured to engage with both the first rail

and the second rail, wherein the drawer comprises one or more tabs extending from the drawer flange on at least one of the first side and second side of the drawer, each tab extending from the drawer flange configured to engage either the first slot defined in the first rail flange of the first rail or the second slot defined in the second rail flange of the second rail.

16. The refrigerator appliance of claim 15, wherein the one or more tabs of the drawer comprises at least three tabs, the at least three tabs comprising a first tab and a third tab extending from the drawer flange on the first side of the drawer and a second tab extending from the drawer flange on the second side of the drawer.

17. The refrigerator appliance of claim 16, wherein the first tab comprises a first protrusion extending in the lateral direction, the first protrusion configured to engage the first slot of the first rail in the lateral direction.

18. The refrigerator appliance of claim 16, wherein the third tab comprises a second protrusion extending in both the lateral direction and the transverse direction, the second protrusion configured to engage a third slot of the first rail in both the lateral direction and the transverse direction.

19. The refrigerator appliance of claim 16, wherein the second tab comprises a third protrusion extending in the transverse direction, the third protrusion configured to engage the second slot of the second rail in the transverse direction.

20. The refrigerator appliance of claim 15, wherein each tab of the one or more tabs extending from the drawer flange comprises a snap configured to engage, in the vertical direction, a bottom side of one of the first rail flange or the second rail flange.
